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On the origins of dryland vegetation patterns

A first attempt to grow
vegetation patterns in the lab
with the clonal grass *Paspalum
vaginatum* L.

Enrico Magazzino¹, Angeles Garcia
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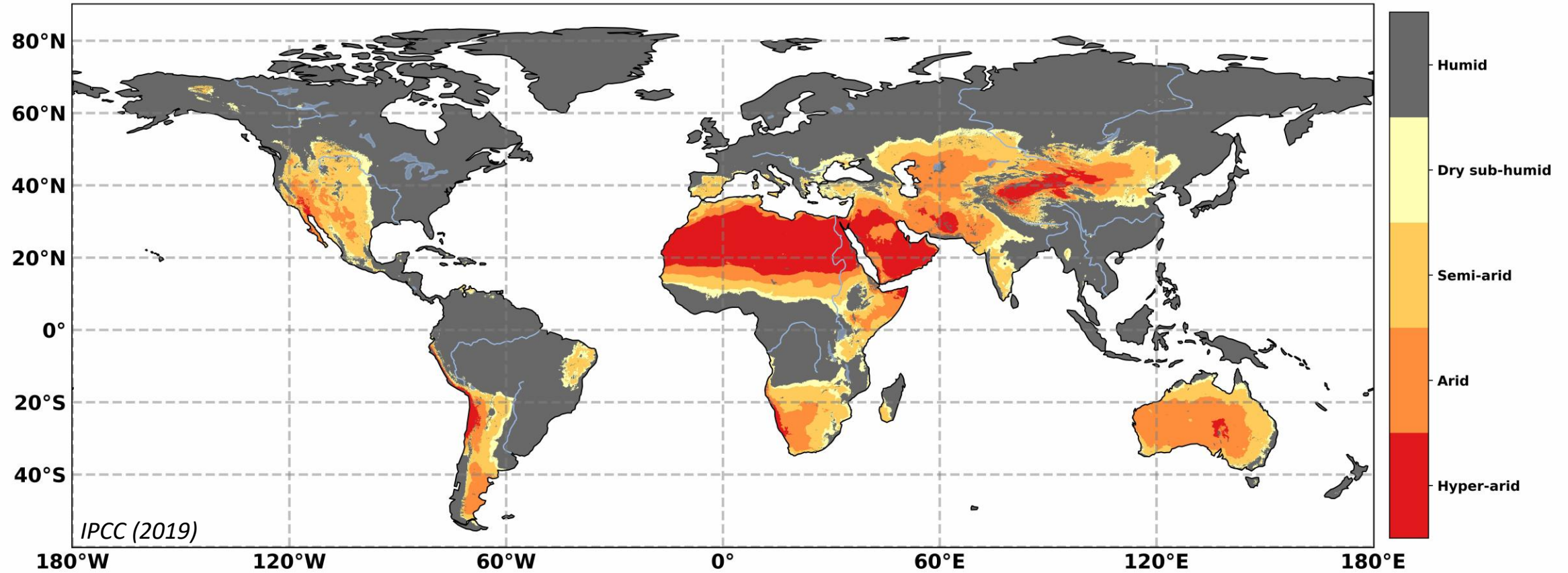
NOW Life conference, May 23th 2023



Open Competition Domain Science – XS

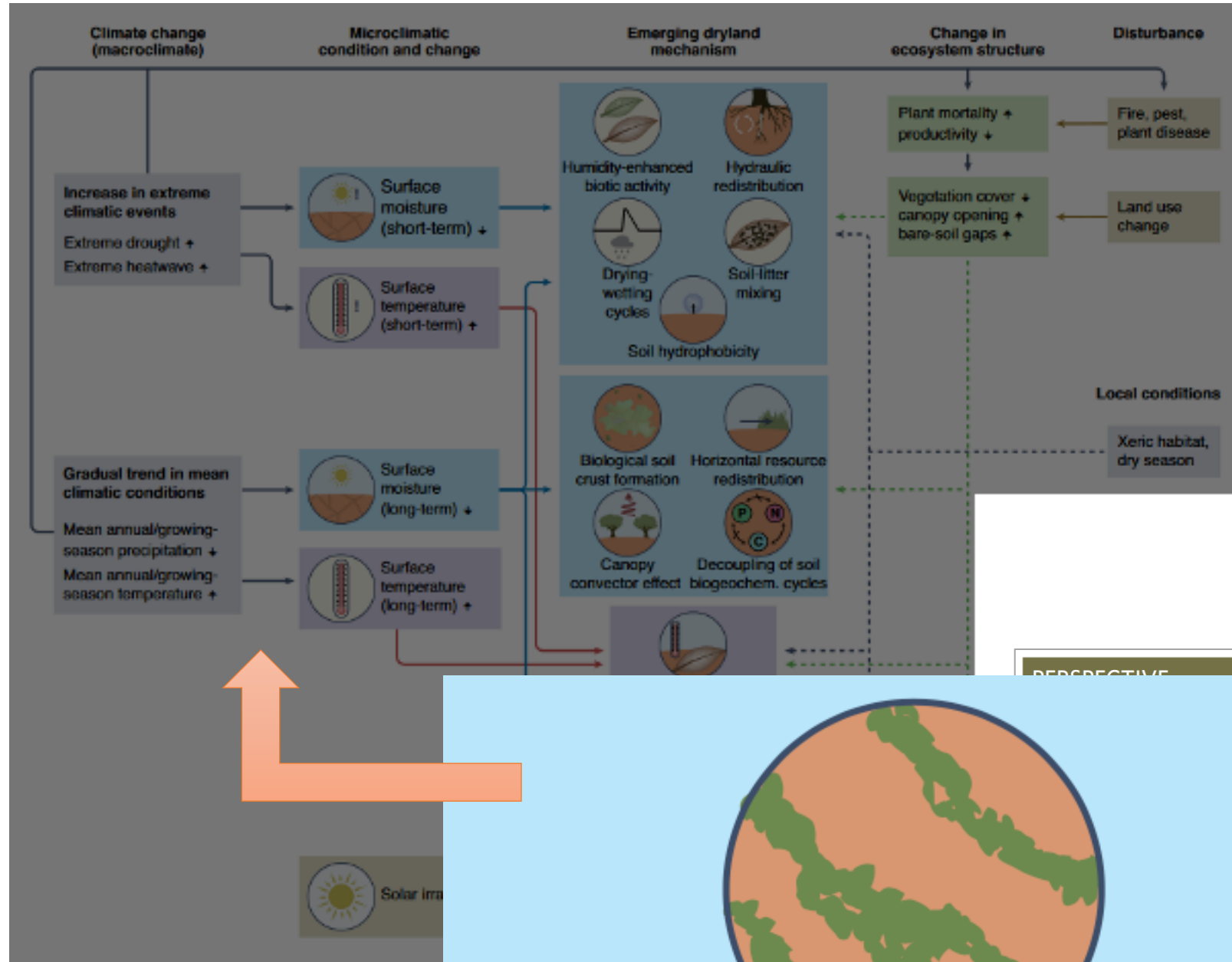
The Open Competition Domain Science – XS grants of a maximum of € 50,000 are intended to support promising ideas and to facilitate innovative and more speculative initiatives within the seven Domain Science disciplines. The proposed research is ground-breaking and high-risk. What counts is that all results, be they positive or negative, must contribute to the advancement of science.

World Drylands



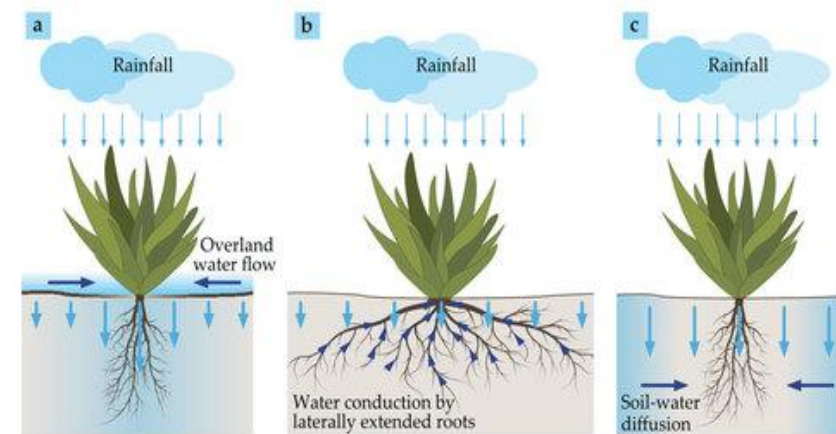
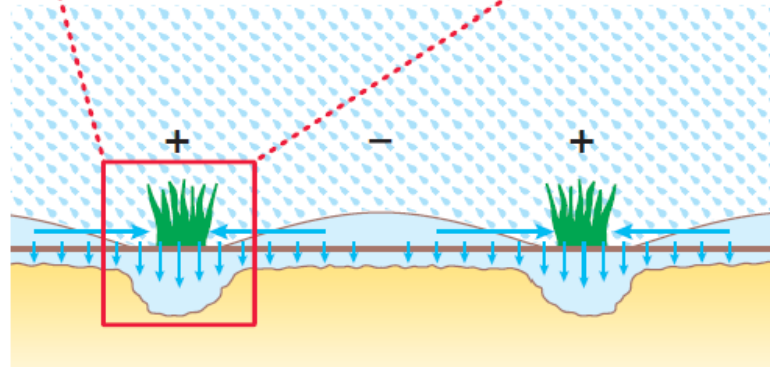
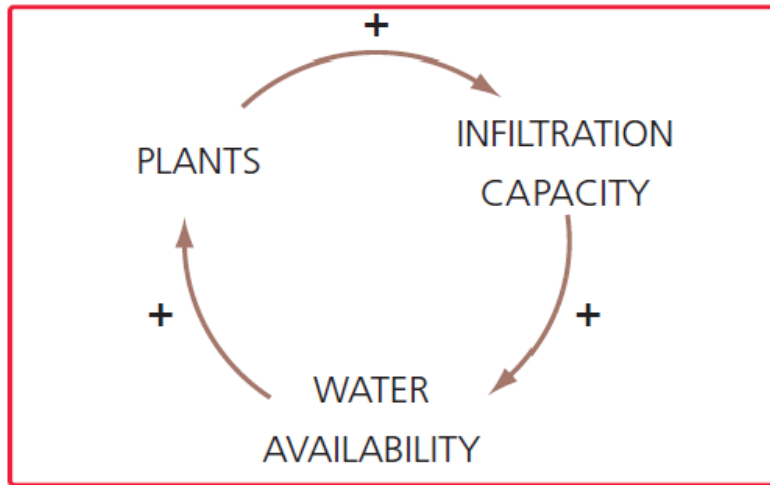
- Annual precipitation is less than 2/3 of potential evapotranspiration (aridity index of 0.05 to 0.65)
- 46% of Earth's land surface; home to 38% world population (90% in developing countries)
- Up to 56% of Earth's land surface by 2100 (Huang et al 2017)

Increasing importance of dryland mechanisms



Theory on vegetation pattern formation

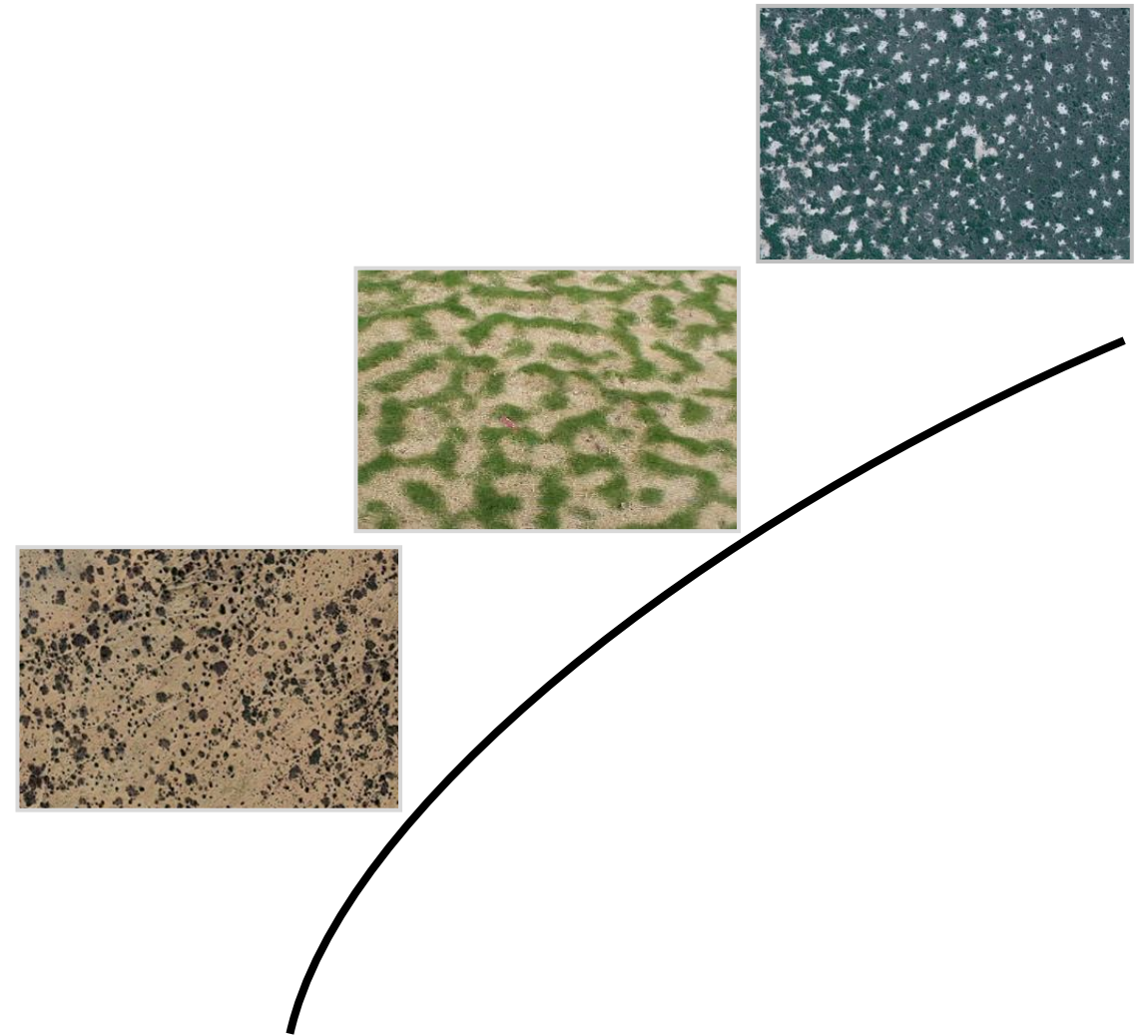
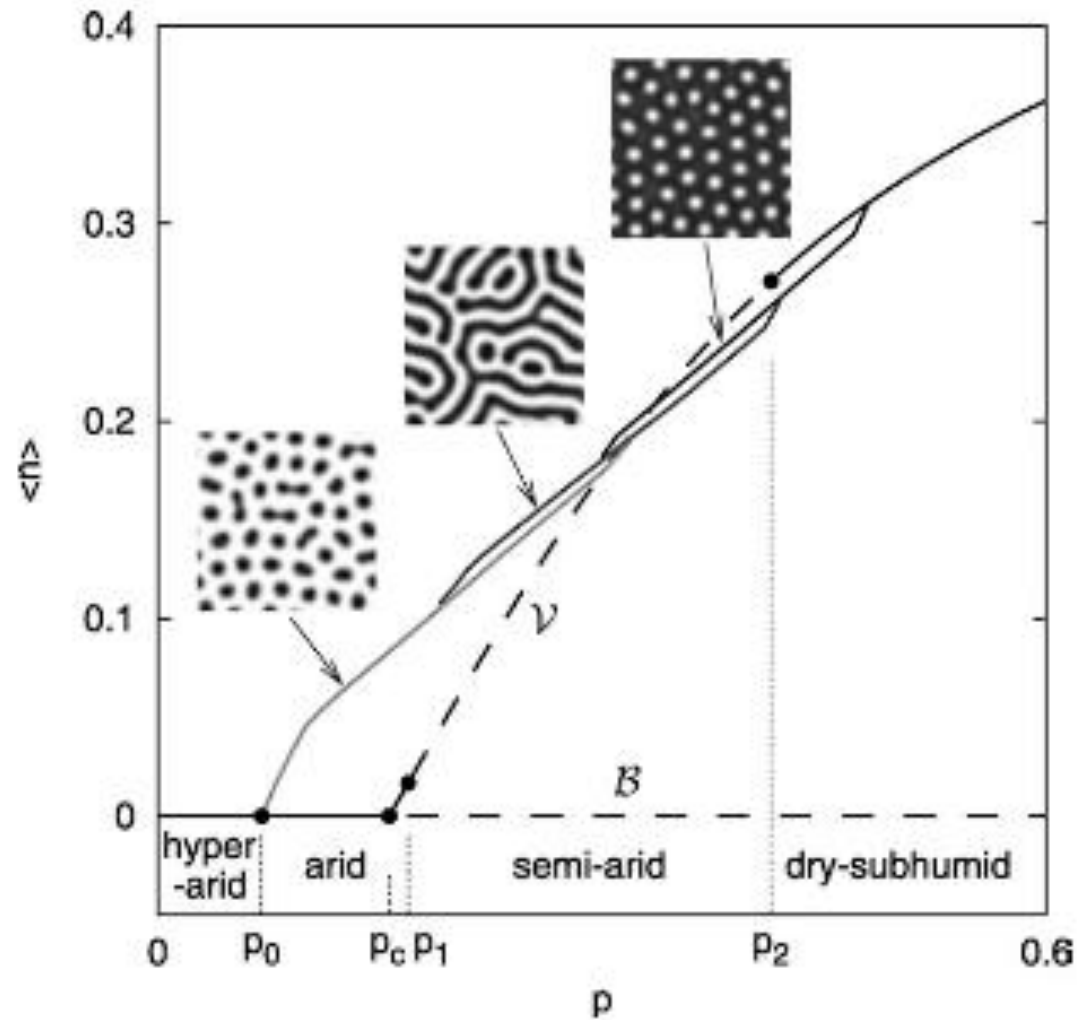
Scale-dependent feedbacks



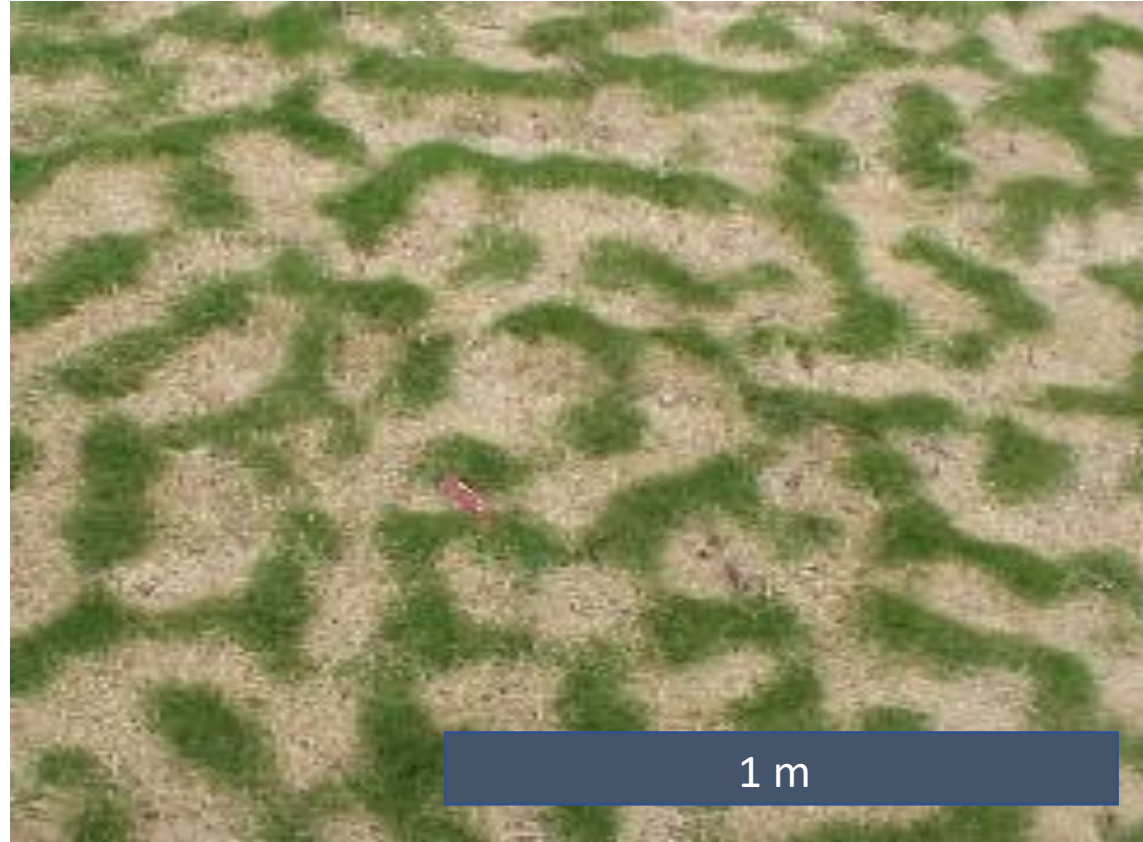
Meron 2019, Physics today.

Theory on vegetation pattern formation

E. Meron et al. / Chaos, Solitons and Fractals 19 (2004) 367–376



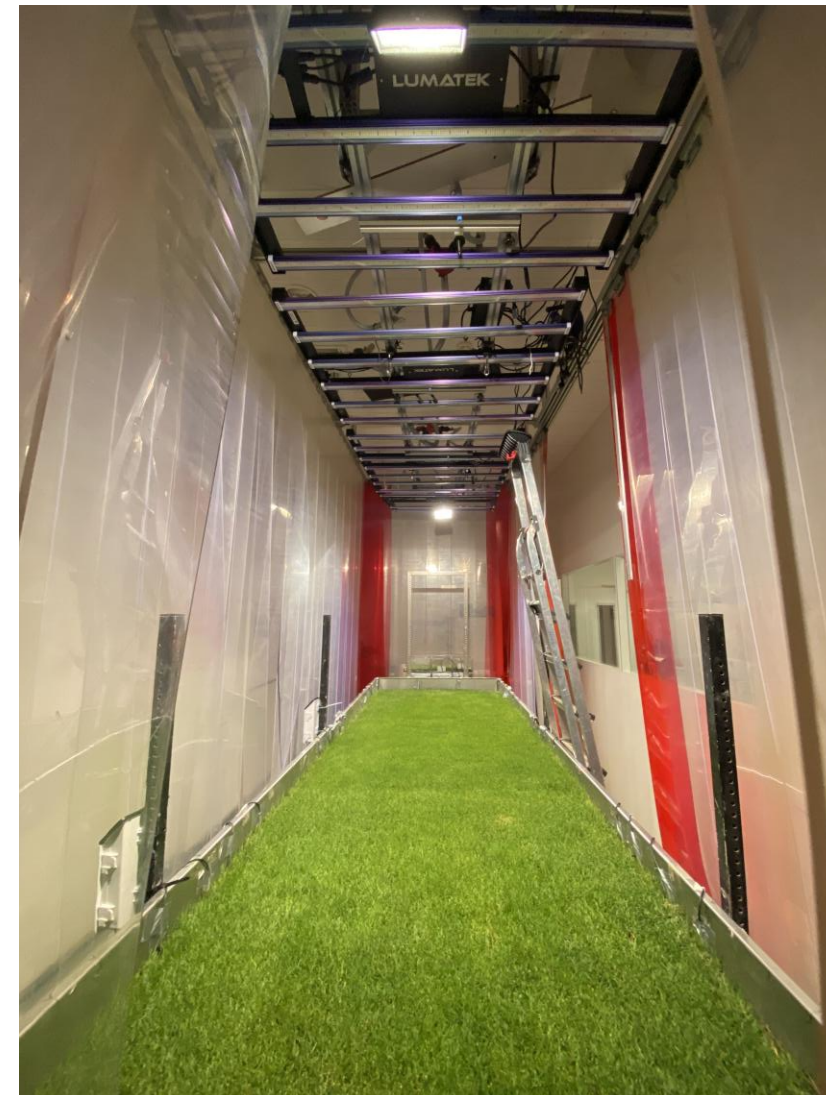
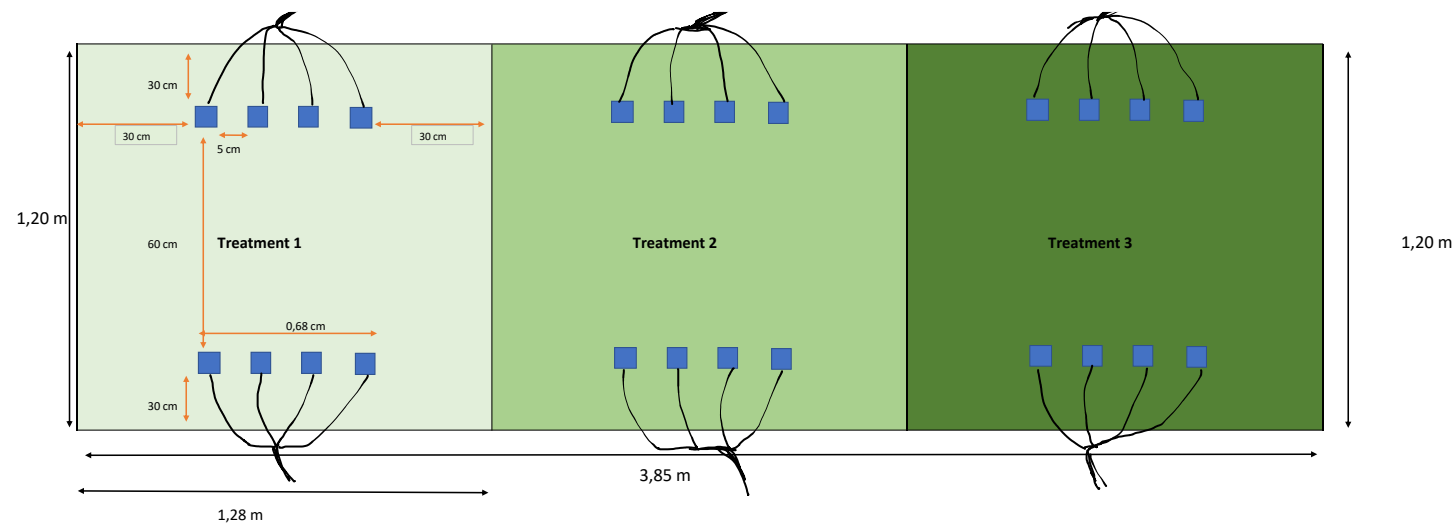
A first attempt to grow vegetation patterns in the lab



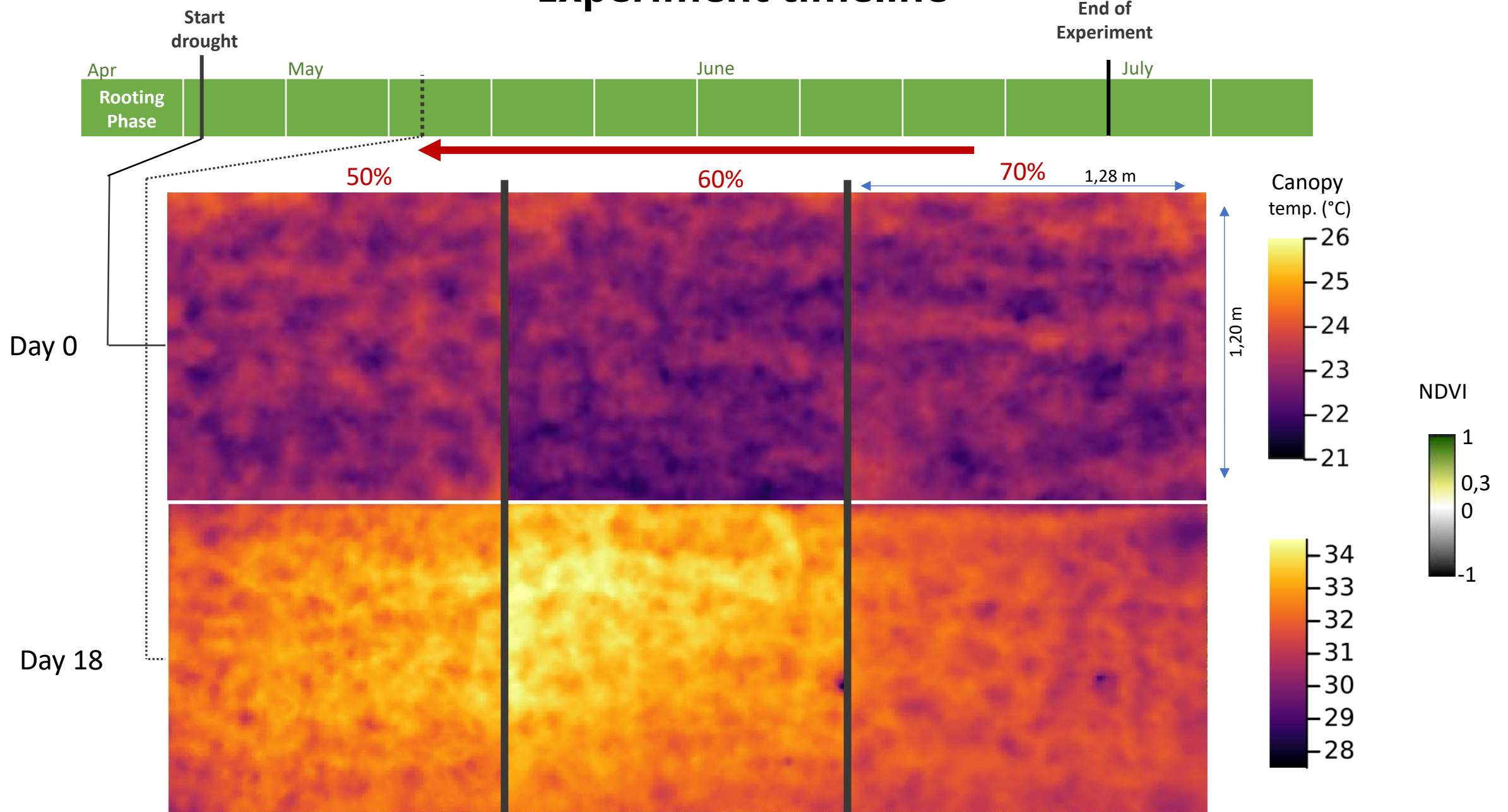
***Paspalum vaginatum* L.**

observed in a residential neighbourhood in the northern Negev
(200 mm mean annual rainfall).

A first attempt to grow vegetation patterns in the lab

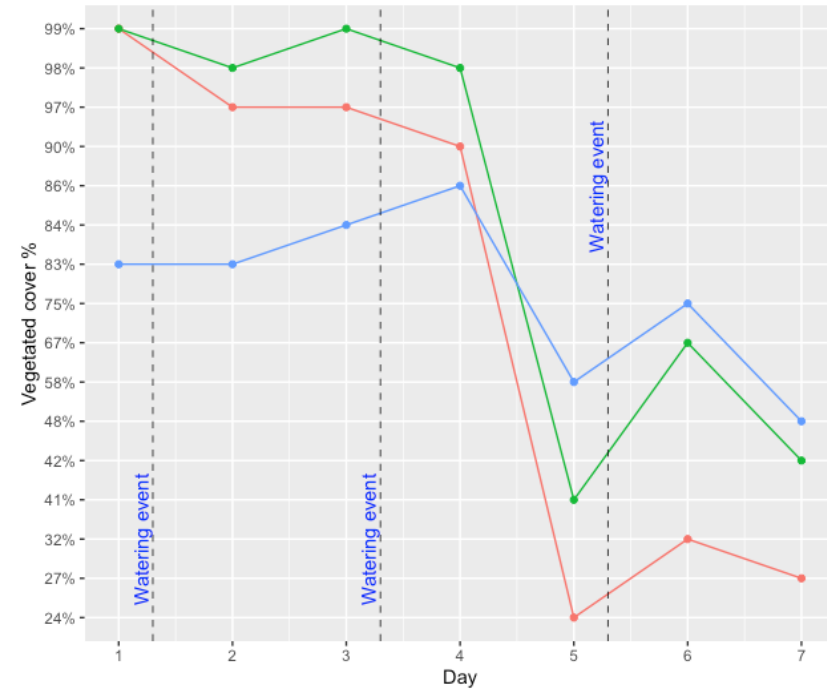


Experiment timeline

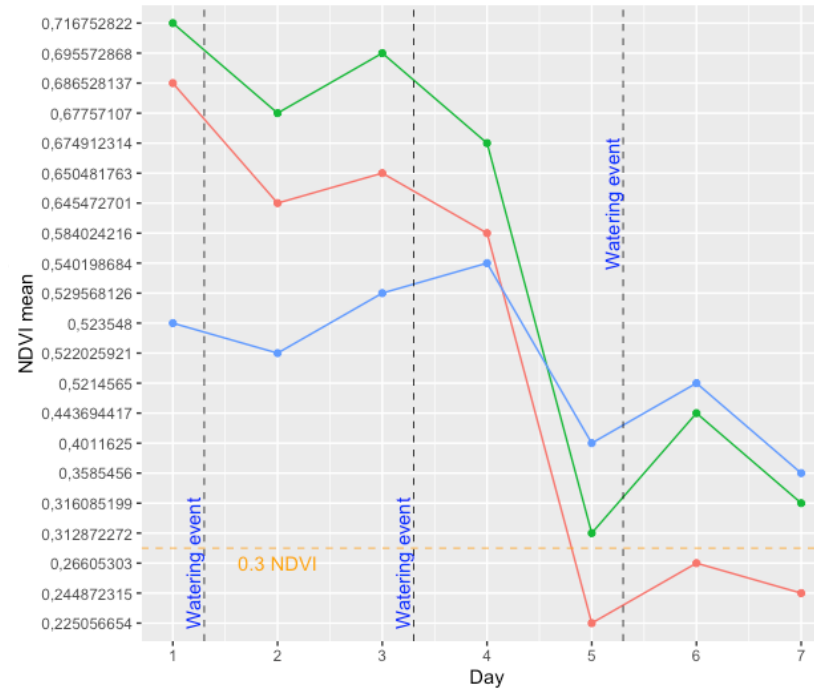


Treatment's changes over time

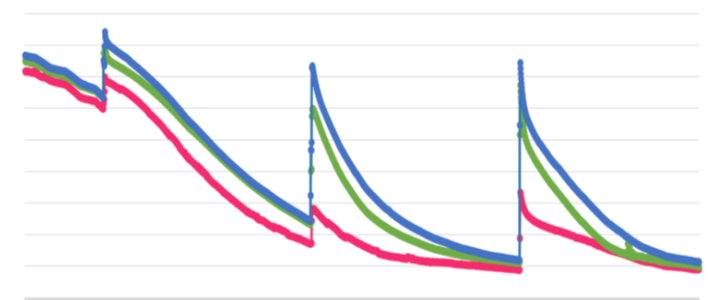
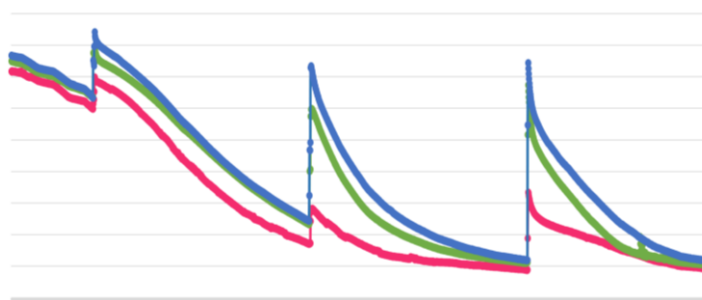
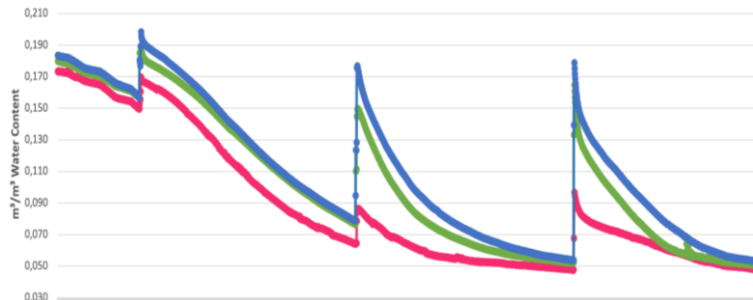
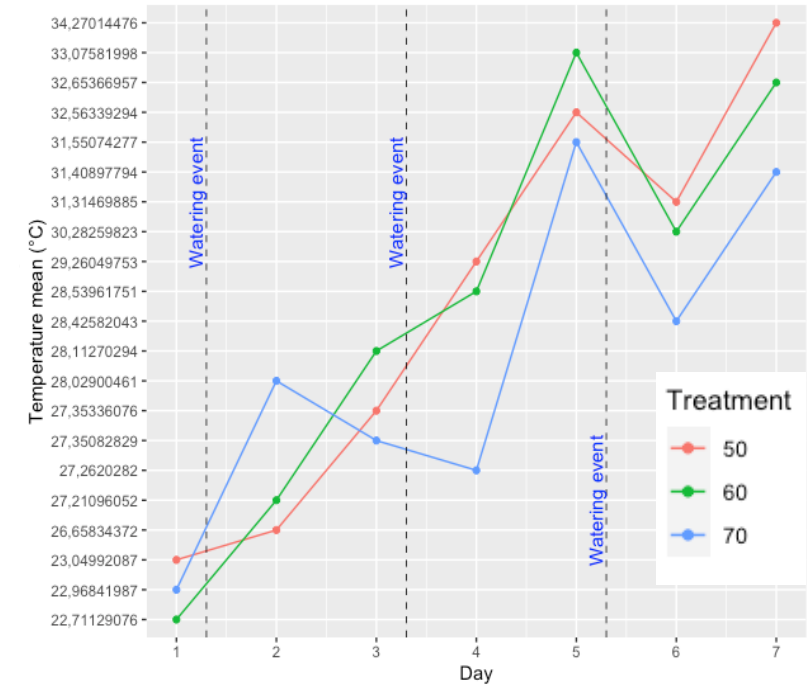
Canopy Cover in time



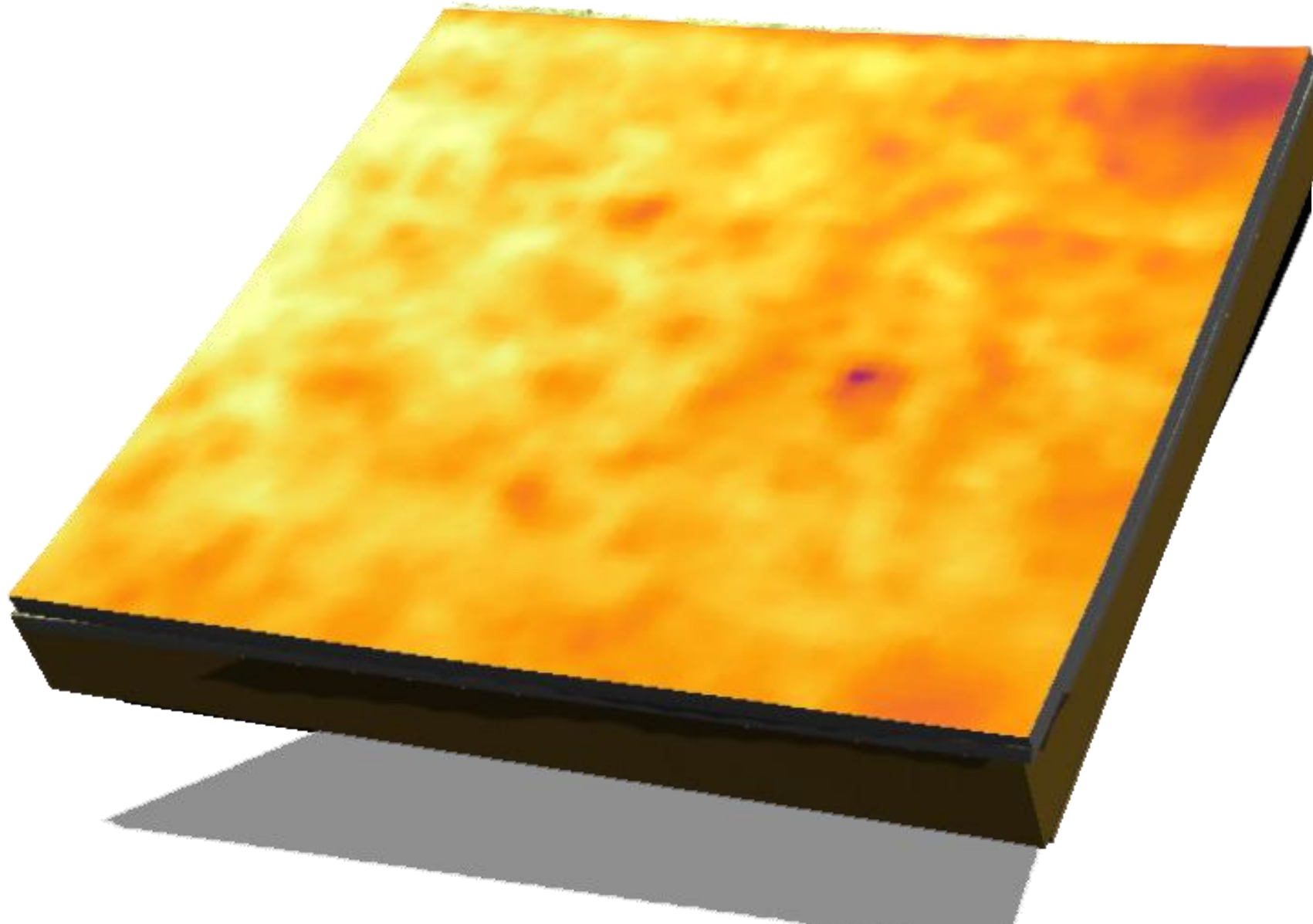
Canopy NDVI in time



Canopy temperature in time

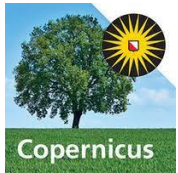


Next step:
Pixel to pixel correlation



Conclusions

- The system needs to reach the stabilization before establishing if there is any pattern formation
- From visual analysis, no regular pattern formation is observed so far
- The experiment is providing parameters for developing guerrilla-like clonal grasses' models and pattern formation in drylands (*Andrea Davin MSc, work presented yesterday during the poster session*)



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Questions?

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